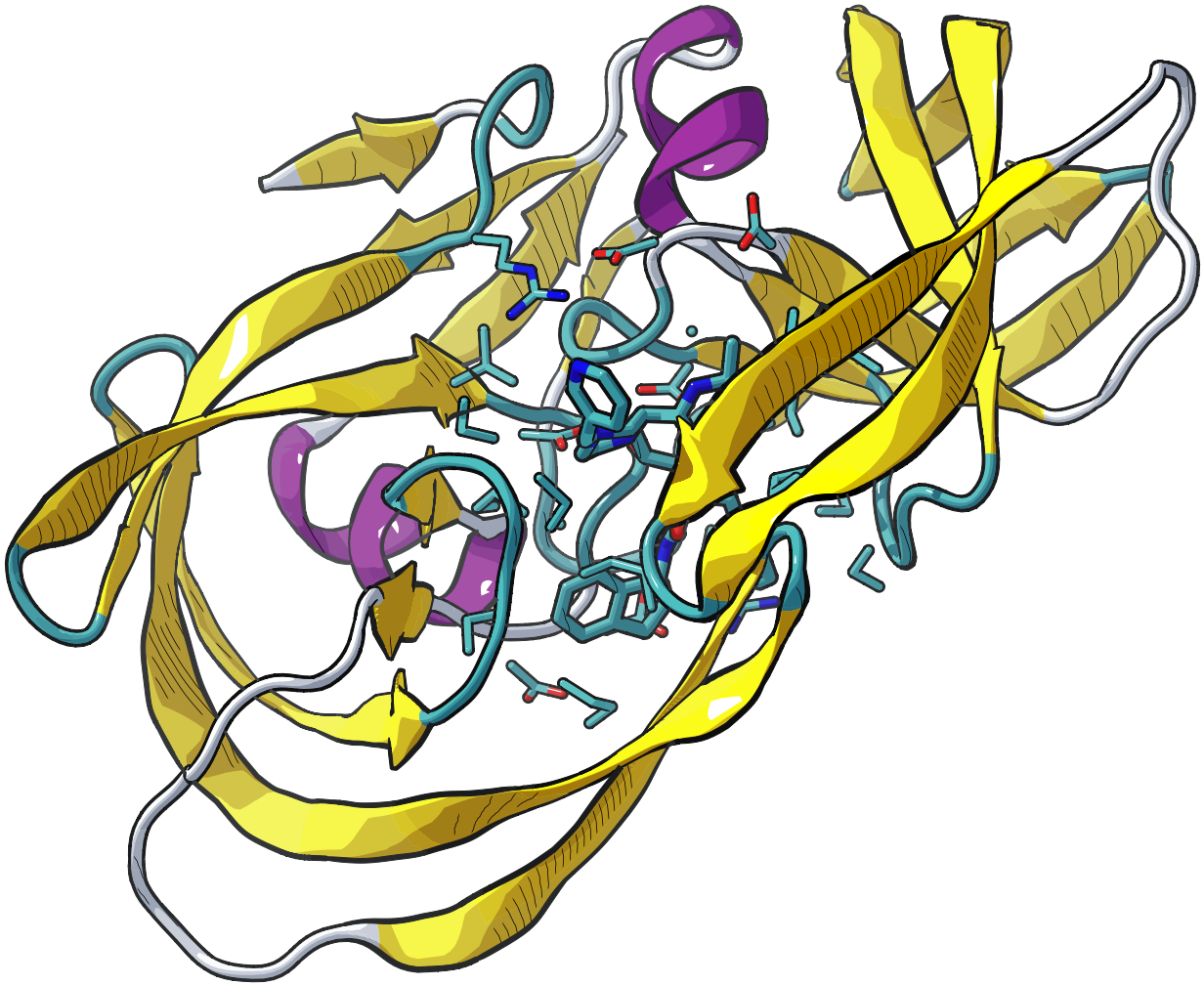
SKIAGRAPH · STYLE BOOK

Comic Ink

Art direction
for Skiagraph



Version 1.0 · stage 5 · 2 November 2026
Public edition: public structures only (RCSB 1HSG, 1UBQ).

HIV-1 protease with the drug indinavir in its pocket (RCSB 1HSG), drawn by Skiagraph in Comic Ink with no options: the default, stage 3.

source: docs/style-book/hero.tcl → skiagraph svg (default; 2.5 s, 1.1 MB)

Draw the molecule the way an inker draws a hero.

A confident outline decides the shape. Two or three hard tones say where the light comes from. A drawn shadow says what sits on what. Nothing is blended to look like a photograph. The picture reads at a glance at a quarter of its size, holds up under a 400 % zoom, and stays honest about the science.

01

Ink decides form.

Silhouettes are confident and weighted; inner lines are quieter; no line ever crosses what is in front of it.

02

Light is drawn, not blended.

Hard tone bands from one key light at the upper left, computed on the 3D shapes. Highlights are shapes that follow the form, never glows or polygons.

03

Depth through weight, not fog.

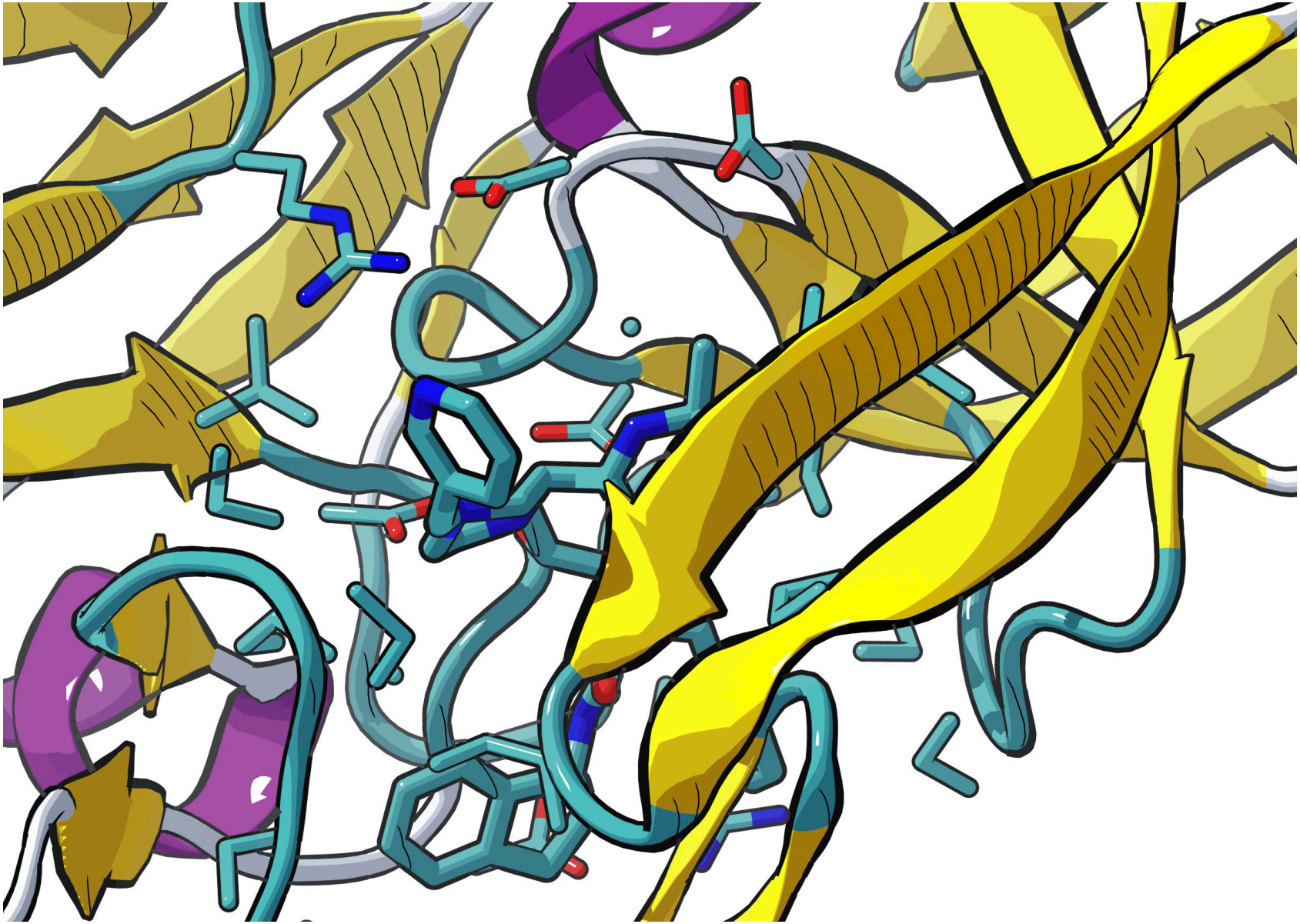
Far shapes get thinner, lighter lines and calmer tones. They never fade away.

04

Colour stays honest.

Element and structure colours stay recognisable in every band. Ink is near-black and never takes the fill's colour.

For scientists making figures for papers, posters and talks: a plain SVG in which every outline, tone, shadow and hatch is a shape that can be selected and changed in Inkscape, and that prints sharp at any size.



The drug in its pocket: indinavir (sticks, carbon cyan, nitrogen blue, oxygen red) among the side chains of HIV-1 protease that hold it (RCSB 1HSG). Shown at about 3× the cover's scale. Stage 3 default: one continuous line per edge, boldest around each object; tone bands, cast shadows and hatching from a key light at the upper left, all computed on the 3D shapes.

source: cover render, cropped, drawn by resvg at 2400 px wide

Four traditions, no borrowed images.

Described in words only. Every picture in this book is Skiagraph's own render; no third-party image is shown or copied.

Graphic-novel ink

Weighted contours, spotted blacks, a line that thins as the form turns away from the light.

We take: Line weight that means something.

Cel-shaded games

Two or three flat tone bands with hard edges, outlines found from breaks in depth and surface direction.

We take: Hard bands from real light; outlines from depth jumps.

Technical illustration

Clarity before realism: one key light from the upper left, a strict line hierarchy (silhouette, crease, detail).

We take: One key light; silhouette › crease › detail.

Scientific illustration

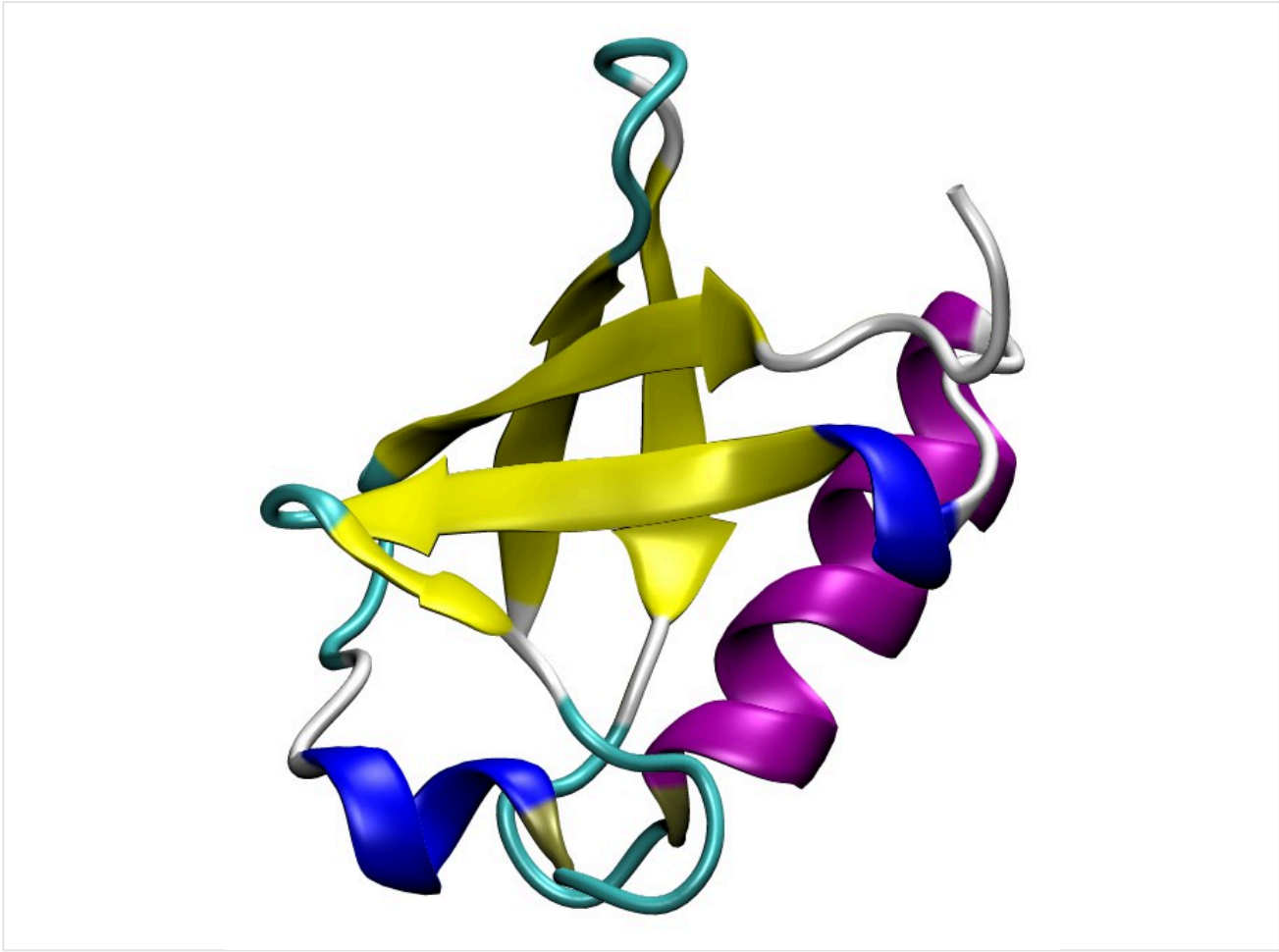
David Goodsell's watercolours: flat colour and a clean outline keep molecules legible even in a crowded cell. Irving Geis's ink-and-airbrush protein portraits: structure first, light used to separate layers.

We take: Legibility in crowds; structure colours first; light separates layers.

The photographic look.

Skiagraph began by copying VMD's ray-traced picture exactly, with thousands of fitted gradients. It was faithful and heavy, and it looked like a photograph, not a drawing.

Also out: glows, airbrushed gradients that imitate a renderer, fog that erases what is far away, and scratchy texture that adds noise rather than form.



Ubiquitin (RCSB 1UBQ) drawn by the old VMD-matching renderer: 13.8 MB and 295 000 elements for one cartoon. The same scene in Comic Ink is 0.47 MB.

source: skiagraph svg (code of 96d1006, before stage 1) on tests/matrix tri_newcartoon, 1200×912

3 · EXPLORATION · STAGE 0

Three directions, one frame.



A · Comic Ink
`--style comic-ink (the default)`

CHOSEN

Strongest at 25 %: confident silhouettes, saturated ligand colours, a drawn shadow that separates layers. Weak: lines far too heavy on large shapes, tone bands that barely show, a sticker-like offset shadow.



B · Clean Line
`--style fill=pale:0.12,lines=black,w=1.2,…`

KEPT IN PART

Calm and legible, but flat: the helices lose their volume. We keep its treatment of far shapes: thinner, lighter lines, never faded.



C · Engraved Print
`--style fill=pale:0.45,hatch=engrave,bg=#f7efdf,…`

KEPT IN PART

A print feel, but low contrast and muddy at 25 %. Its two-ink version erased the element colours. We keep only hatching in the darkest band, drawn across the form.

WHY A WON

The only direction that reads at a glance at a quarter of its size and still looks drawn at 400 %.

It is also the closest to the brief: ink outlines, hard tones, drawn shadows. Clean Line explains as well at full size but loses volume; the Engraved Print turns to mud when small.

Decision by the owner after the stage 0 review, 1 November 2026, with an art director: A is the base; B gives the calm far shapes; C gives only the dark-band hatching. The "latest recipe" of October is abandoned.

READING TEST

50 %



A B C

25 %



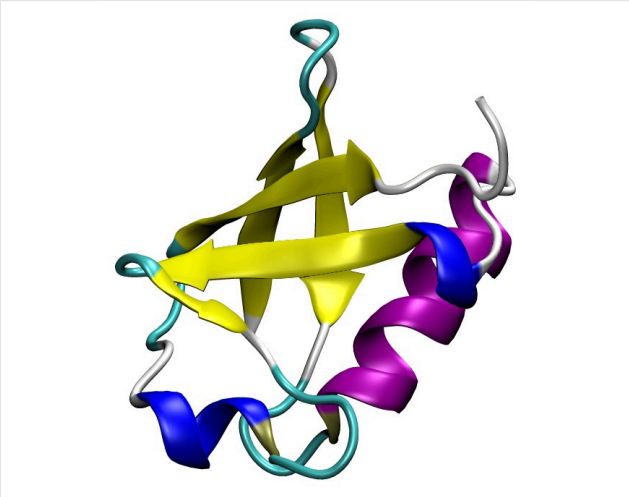
A B C

The same three crops at half and a quarter of the width shown opposite. At 25 % only A still separates the ligand from the protein.

From photograph to drawing.

2026-10-11 → 14

Match VMD



Reproduce VMD's ray-traced picture exactly with fitted gradients. Faithful, photographic, heavy: 62 MB and 1.27 million elements for the cover frame.

source: old renderer (code of 96d1006) · tri_newcartoon, 1UBQ

2026-10-13

Flat prototype



One flat colour per shape, darker with depth; lines only at depth jumps. Tiny (0.6 MB) and fast, but no light at all.

source: --style flat · cover frame, 1HSG

2026-10-15

Style laboratory



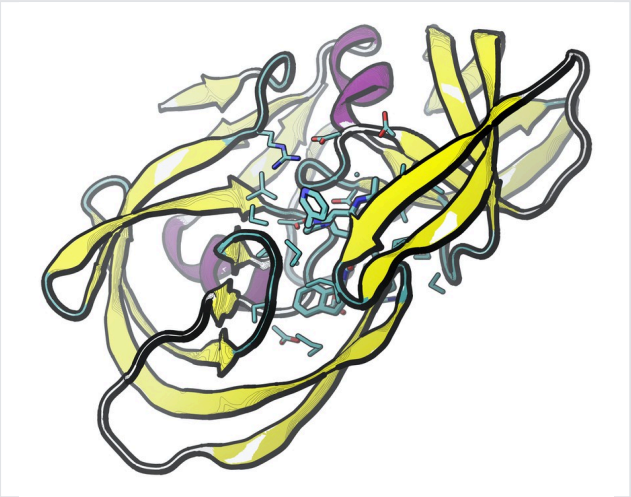
Cel bands, hatching, airbrush, duotone, rim light and drop shadow as recipes; 26 combinations compared side by side.

source: --style fill=cel:3,hatch=lines,fog=0.8,… · 1HSG

Every picture up to stage 1 was re-rendered on 1 November 2026 with the code of its time (the old renderer from commit 96d1006; the styles from the stage 0 code), on public structures. Later stages add their steps here.

2026-10-15

Refinements



Engraving lines, light-weighted ink, shaded sticks, tinted fog, glints. Abandoned: far ribbons wash out and engraving reads as scratches.

source: `--style …,hatch=engrave,…,glint=protein · 1HSG`

2026-11-01

Stage 0 · three directions

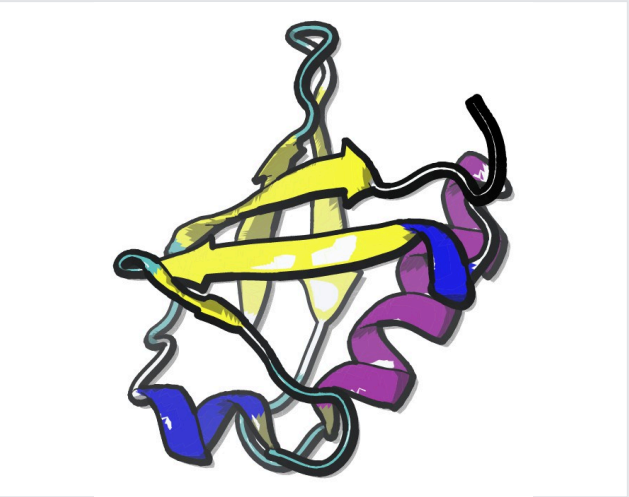


Comic Ink chosen as the base, with Clean Line's calm far shapes and the Engraved Print's dark-band hatching.

source: `--style comic-ink · 1HSG`

2026-11-01

Stage 1 · foundation



No change to the look: the 126 stage 0 renders re-made byte for byte after the refactor (the 127th is the same refusal as before). Comic Ink is the default; the same cartoon is now 0.47 MB.

source: `skiagraph svg (default) · tri_newcartoon, 1UBQ`

2026-11-02

Stage 2 · drawn light



The first default: lines by the weight rule, tone bands and highlights traced on the visible surface. Lines broke into dashes; coils striped.

source: stage 2 iteration 1 · 1HSG

2026-11-02

Stage 2 · clean outlines



Lines cut only where another part is in front, round ends, ticks dropped by joining line pieces into lines.

source: stage 2 iteration 7 · 1HSG

2026-11-02

Stage 2 · smooth bands



The light smoothed within each part: band edges follow the form, not VMD's flat strand facets.

source: stage 2 iteration 8 · 1HSG

2026-11-02

Stage 2 · the default



Straight hatching in the darkest band only, tapered ends, contact accents; old styles folded into one short recipe.

source: stage 2 final · 1HSG

2026-11-02

Stage 3 · continuous lines

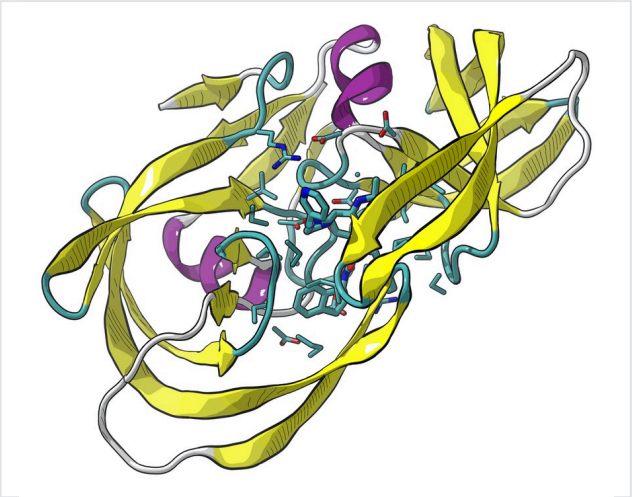


Every edge one continuous line whose width changes smoothly; the outer silhouette 1.75 times bolder. Light still as in stage 2.

source: stage 3 step A · 1HSG

2026-11-02

Stage 3 · light from geometry

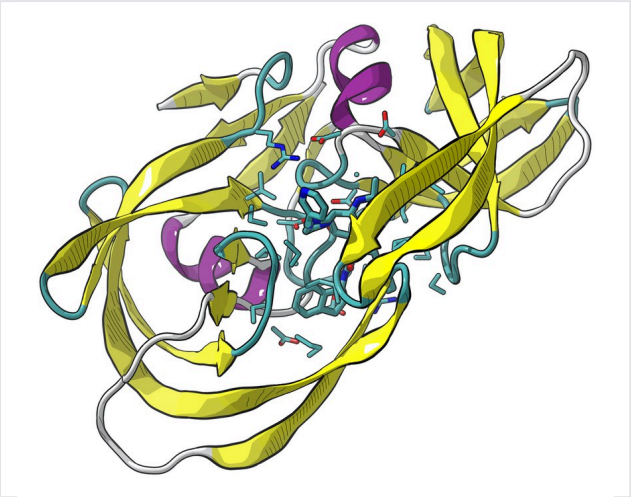


Comic Ink's own key light from the upper left. Bands, shadows, highlights and cross-contour hatching computed on the 3D shapes, never traced from a sampled picture.

source: stage 3 final · 1HSG

2026-11-02

Stage 4 · clean colour cuts, contacts

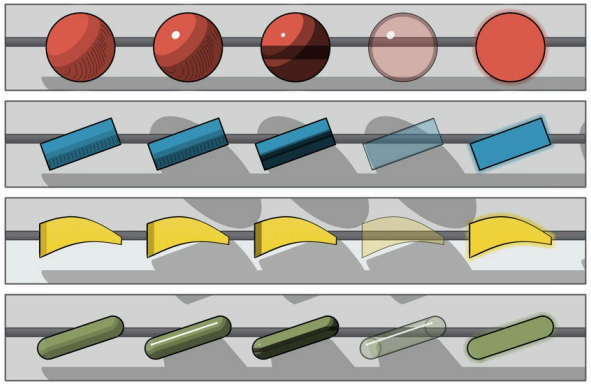


Colour changes along a strand or coil cut cleanly across it (split at the middle between VMD's rings of points); fewer strokes on small and far shapes; contact accents and shadows on balls rebuilt from the geometry.

source: stage 4 final · 1HSG, shadows=on

2026-11-02

Stage 4 · materials



Matte, plastic, metal, glass and emissive, read from VMD's material or set per group in the recipe; each changes which light marks a shape gets.

source: stage 4 material chart · made-up geometry

2026-11-02

Stage 4 · lighting presets

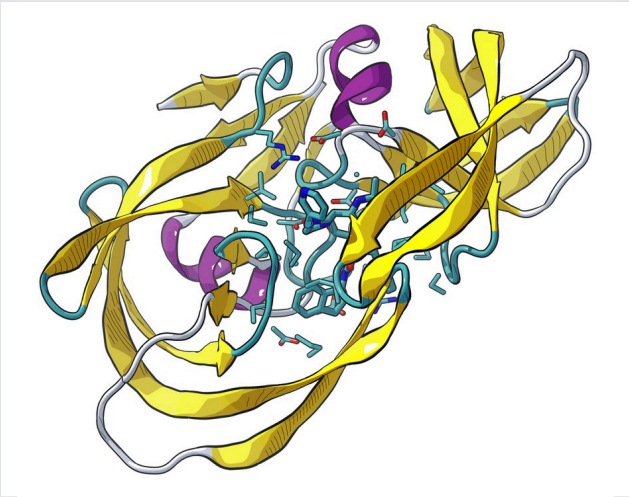


Studio (the default), rim, low, night and VMD's own light; VMD's shadow switch is honoured. Shown: rim light.

source: --style light=rim,shadows=on · 1HSG

2026-11-02

Stage 5 · shading in colour

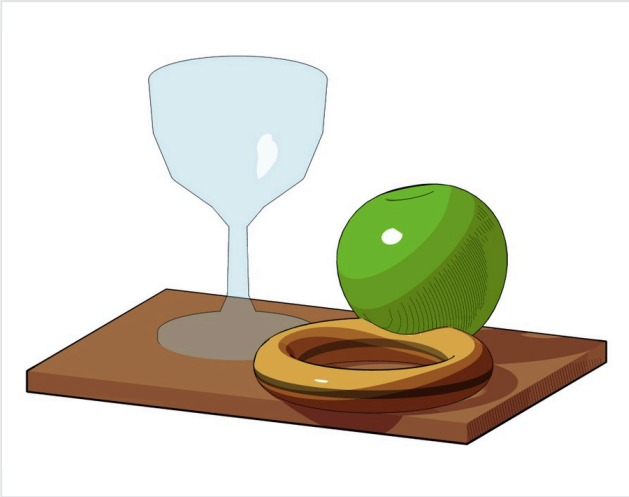


Each tone band a deeper, hue-shifted version of the colour under it instead of black ink over it: yellow shades to golden orange, not olive. Cast shadows on by default.

source: 1.0.0-rc1 default · 1HSG

2026-11-02

Stage 5 · beyond molecules

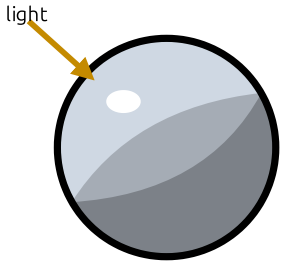


OBJ and glTF models read into the same scene; their materials mapped onto the five; a camera that frames the model. The same Comic Ink, no molecule-specific step.

source: examples/meshes/still_life.glb · default

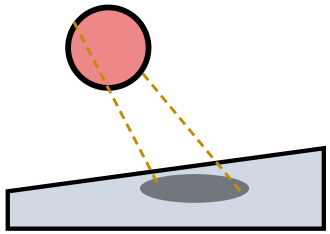
Every shape comes from the 3D shapes.

Skiagraph never paints a picture and traces it. It knows each sphere, tube and triangle in the scene, and it computes where each line, band, shadow and stroke lies on them, then projects that onto the page. That is why the drawing stays sharp at any size and every piece can be selected and edited.



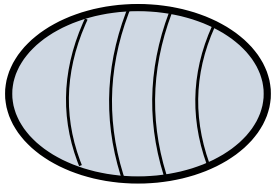
Bands

The light falls off across a ball. Where it crosses a band's level is a circle on the ball; on the page, an ellipse. On a tube it is two straight lines along it; in a mesh triangle a short curve, computed to 1/20 px.



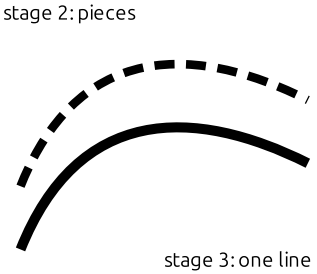
Cast shadows

The outline of a shape as the light sees it is projected along the light onto the surface behind. What falls inside is in shadow: an exact shape, not a cloud of samples.



Hatching

The surface is cut by evenly spaced planes, tilted towards the viewer, so each cut curves around the form. Only the parts in the shadow band are kept, and they become tapered strokes.



Lines

Each outline is computed from the exact visible region of every shape. The pieces are then chained into one line per edge, drawn as one filled shape whose width changes smoothly, boldest around each object.

Eye rays and shadow rays are still used, but only to answer yes-or-no questions about the exact shapes while a drawing piece is being built ("is this point covered?", "is this point in shadow?"), never to make a grid of samples that is then outlined. ADR 0015 records the method.

Twice the size, the same drawing.

The same scene converted at 1200 and at 4800 pixels wide. Because every shape is computed from the 3D shapes, not from pixels, the larger file holds the same curves, scaled: shadows and highlights have exactly the same pieces (103 and 73), tones 1324 against 1345, and the pictures differ only by hairlines along the edges (0.14 % of the picture).

Measured by scripts/resolution.py. Right: the tone shapes alone, 1200 px and 4800 px versions drawn at one size, and where they differ.

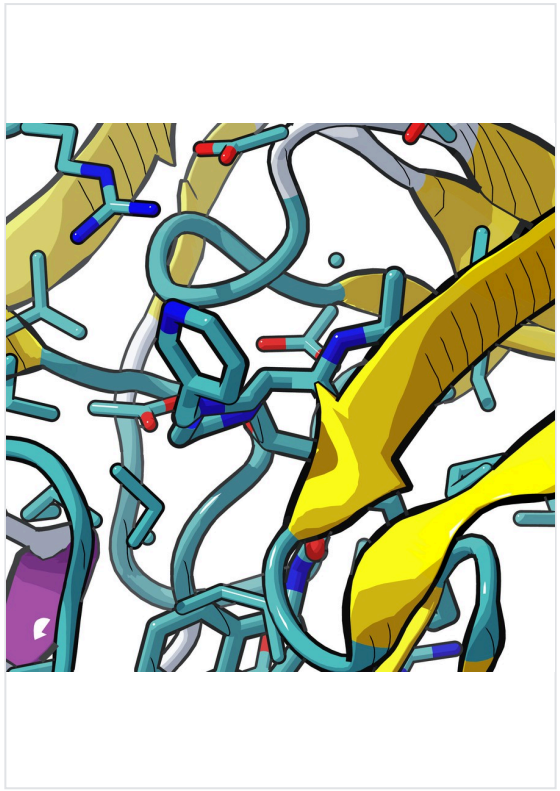


Tone shapes: 1200 px (left), 4800 px (middle), difference (right).
scripts/resolution.py · 1HSG

Each edge is one continuous line. Its weight says how important the edge is: the outline of an object against the background boldest, inner silhouettes next, creases thinnest; the width changes smoothly, never in steps.

OUTER SILHOUETTE	`min(1.75 × inner, max(inner, 0.2 × W × (1 − 0.4 × depth))) px`
INNER SILHOUETTE	`max(2.0, min(0.15 × W, 4.5) × (1 − 0.4 × depth)) px`
CREASE, CONTACT	`max(1.25, 0.4 × inner) px`
W	the shape's projected width; per 2400 px of picture
DEPTH	0 front … 1 back
AT 25 %	a line keeps ~0.2 of its darkness per px: 1.25 px → 25–29 %, 2 px → 41–47 %, 0.5 px → 10 % (measured, scripts/hairline.py)
CONTINUITY	pieces chained across shapes; width smoothed over 1.5 widths; free ends taper over 3 widths; a line ends only where something covers it
TODAY	one filled shape per edge; 0 visible gaps on the test frames, 1HSG and over the matrix (scripts/defects.py)

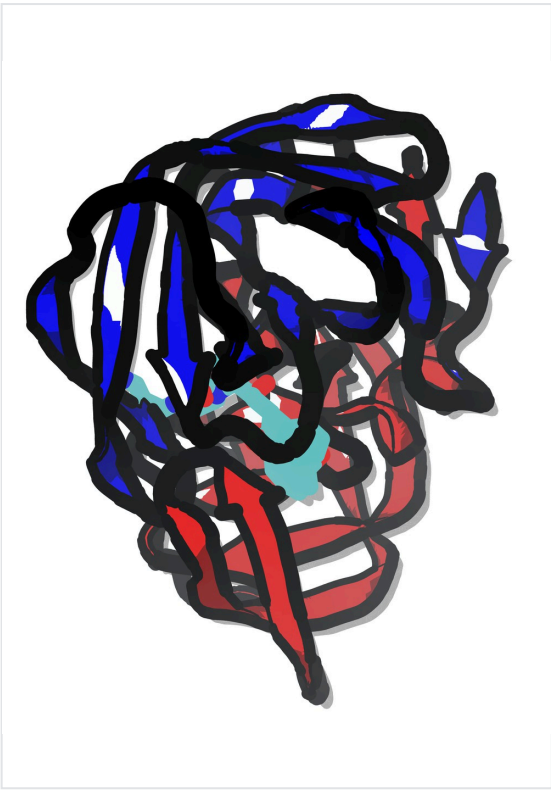
DO



Stage 3: every outline one unbroken line, boldest around each object, thinning smoothly with depth; free ends taper.

4× crop · cover render, default

DON'T



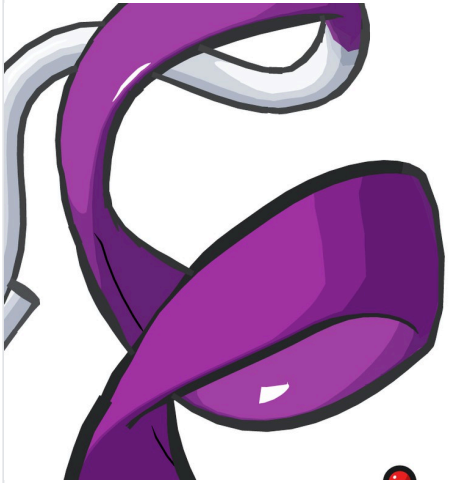
Stage 0's direction A on a small protein: lines as wide as the ribbons swallow the form; helices read as black springs.

4× crop · mix_cartoon_ligand, 1HSG, stage 0 comic-ink

Two hard bands from one key light at the upper left; band edges are curves on the 3D surface, projected: they follow the form, never the triangles. Far tones are calmer.

KEY LIGHT	upper left, 45° up, 45° left, slightly in front (light=scene: VMD's light)
BANDS	light 1.00 · mid 0.78 · shadow 0.56 of the base colour
EDGES	where N·L crosses 0.62 and 0.30: a circle on a sphere, two lines on a tube, a refined curve in a triangle (≤ 0.05 px off)
FAR TONES	band opacity × 1, 0.8, 0.6 over the depth thirds
SPOT BLACKS	spot=1: the core below N·L 0.10 and cast shadows at 85 % ink

DO



Two hard bands from the upper-left key light on a helix and a strand; their edges are curves computed on the 3D surface.

4× crop · cover render, default

DON'T



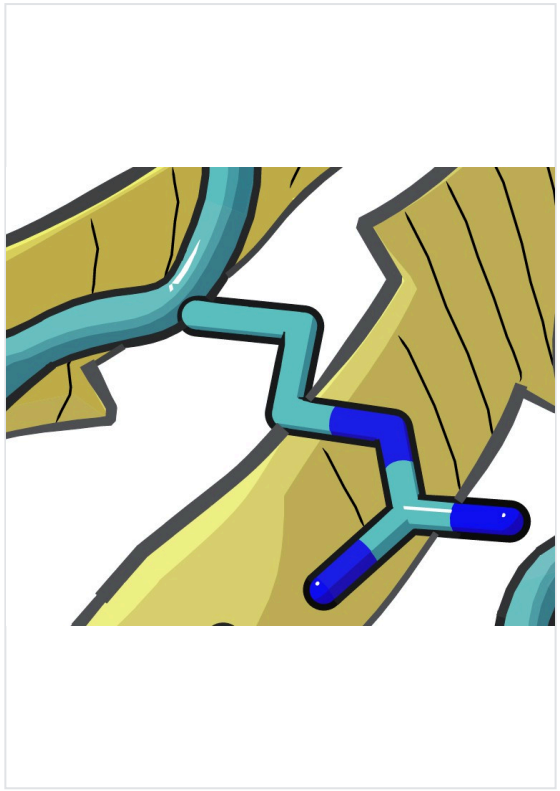
Stair-stepped band edge and jagged polygon glints on a surface: the bands follow the triangles.

4× crop · tri_quicksurf, 1UBQ, comic-ink

Shadows are drawn shapes that fall on what lies beneath, away from the light: the occluder's outline, as the light sees it, cast onto the receiving surface.

CAST SHADOW	occluder outline (circle, tube ends, triangle) cast along the light onto the receiver, clipped to its lit part
NEVER	an offset copy of the shape (a sticker)
CONTACT ACCENTS	stage 4: where a ball or tube of another rep comes within 10 px (in space) of a mesh, the mesh's plane cut through the grown shape, cut to what shows: a dark rim along the resting edge
SWITCH	drawn when VMD's shadows are on (Tachyon Shader_Mode Full); shadows=on|off overrides
TODAY	exact on triangles and (stage 4) on balls; on tubes partial (see notes)

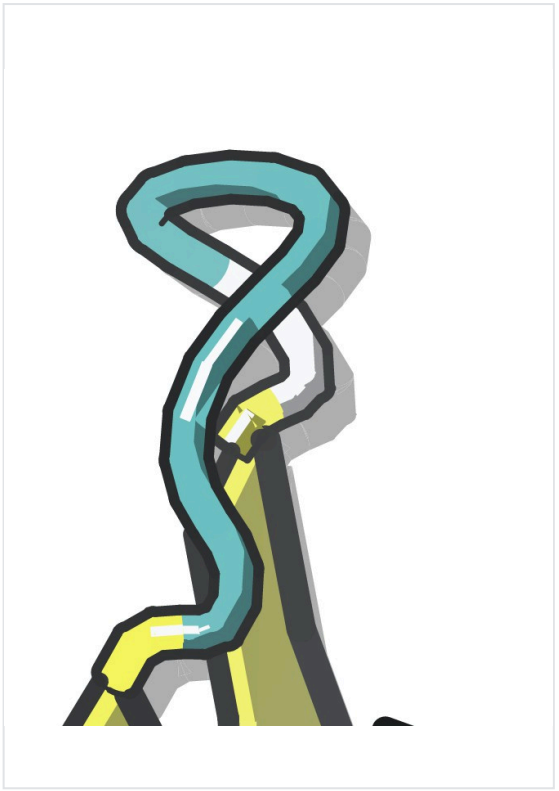
DO



Side chains cast shadows on the strand behind them, down and to the right, away from the upper-left light.

4× crop · cover render, default

DON'T



The drop shadow is a grey copy shifted down and right: it floats beside the coil like a sticker instead of falling on what is behind.

4× crop · style_cartoon, 1UBQ, comic-ink

Hatching in the shadow band, strokes curving around the form (cross-contour); on round forms with no shadow band, lighter strokes in the darker half of the mid band.

STROKES	sections of the surface by parallel planes tilted 35° towards the eye: their pictures curve around the form
SPACING	12 px per 2400 px of picture; every second stroke starts deeper in the band
STROKE	0.9 px, tapered ends, chained across triangles; ≥ 24 px long
MID BAND	spheres and tubes only, 0.6 × width, 70 % ink

DO



Strokes curving across the strands' shadow sides, laid on the 3D surface and clipped to the shadow band.

4× crop · cover render, default

DON'T



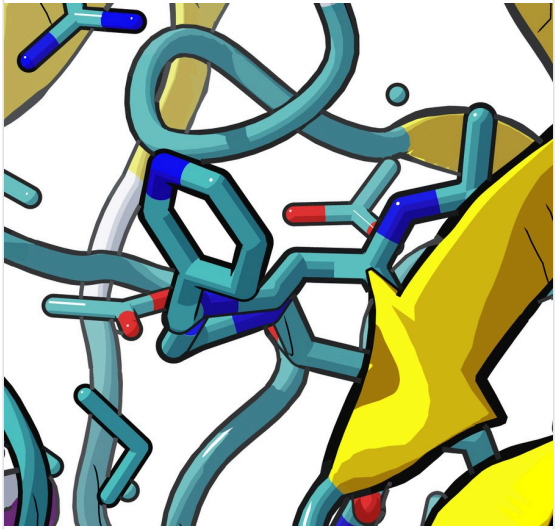
Engraving that follows equal light reads as wood grain and scratches, densest where the eye looks.

4× crop · tri_newcartoon, 1UBQ, latest recipe

Element and structure colours stay recognisable in every band; the ink is near-black and never takes the fill's colour.

FILLS	VMD's colours (page 'Palette'), banded by light
INK	#000000: crisper than a warm near-black on element colours (stage 2)
NEVER	a palette that erases element identity

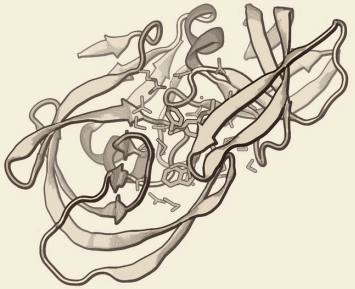
DO



Nitrogen blue, oxygen red and carbon cyan stay clear on the ligand and side chains.

4× crop · cover render

DON'T



Two-ink duotone: nitrogen, oxygen and carbon can no longer be told apart.

C-duo · 1HSG, fill=duo:#2a1d16:#f4e8d0

Depth is told by line weight and value, never by fading to white: far shapes keep a thinner line and a calmer tone.

FOG	at most 0.4, towards #e4eaf1
LINES	× 0.6 at the back, never below the floors (2.0 / 1.25 px)
TODAY	fog 0.4 towards #e4eaf1, colour 40 % greyed at the back; lines × (1 − 0.4 depth) and lighter; band opacity × 0.8, 0.6 in the middle and back thirds

DO



The stage 3 default: far strands keep a thinner, lighter outline and calmer bands, never washed out.
default · 1HSG

DON'T



The latest recipe: the back of the protein fades to near white and loses its outline.
latest recipe · 1HSG

VMD's colours, shaded in colour.

Elements (VMD colouring "Name")

	LIGHT	MID	SHADOW	INK
C carbon VMD cyan	#40bfbf	#2d929e	#1d697c	#000000
N nitrogen VMD blue	#0000ff	#1200d2	#0e00a5	#000000
O oxygen VMD red	#ff0000	#d20000	#a50000	#000000
S sulphur VMD yellow	#ffff00	#d2b600	#a57900	#000000
P phosphorus VMD tan	#808033	#696024	#534317	#000000
H hydrogen VMD white	#ffffff	#c1c7d2	#8b94a5	#000000

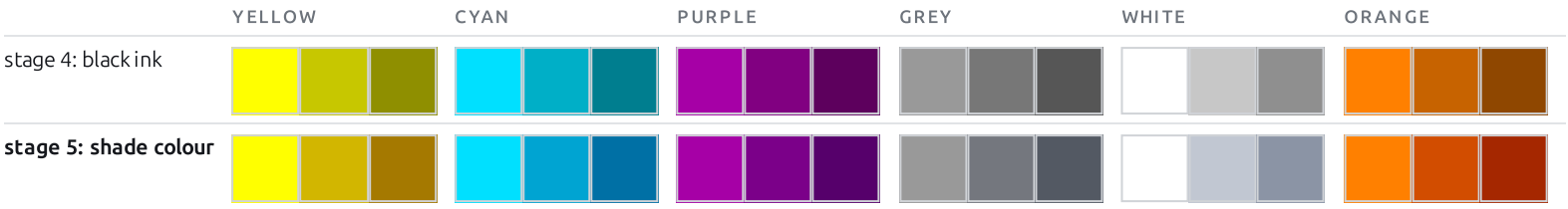
Secondary structure (VMD colouring "Structure")

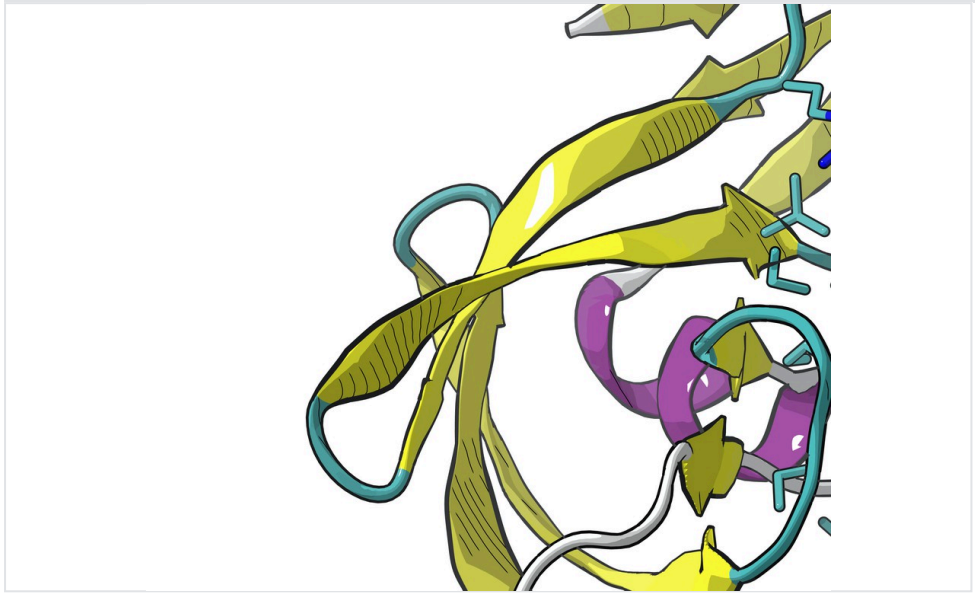
	LIGHT	MID	SHADOW	INK
α helix VMD purple	#a600a6	#7b0089	#56006b	#000000
3_{10} helix VMD blue	#0000ff	#1200d2	#0e00a5	#000000
π helix VMD red	#ff0000	#d20000	#a50000	#000000
β strand VMD yellow	#ffff00	#d2b600	#a57900	#000000
bridge VMD tan	#808033	#696024	#534317	#000000
turn VMD cyan	#40bfbf	#2d929e	#1d697c	#000000
coil VMD white	#ffffff	#c1c7d2	#8b94a5	#000000

Base colours are VMD 1.9.4a57's defaults (read with `colorinfo rgb`). Each colour has a tone ramp: light (the base), mid and shadow. Since stage 5 the bands are solid colours made from the base, as a comic colourist shades: darker (value × 0.82, then × 0.65), more saturated, and turned a little in hue: warm colours towards red (yellow to golden orange), cool ones towards blue-violet (cyan to deep teal, purple to deep violet), greys towards cool slate. Black ink at an opacity, used until stage 4, turned yellow olive and greyed the whole picture; black is now kept for lines, hatching and spot blacks. White coil and hydrogen keep their outline to stay visible on white.

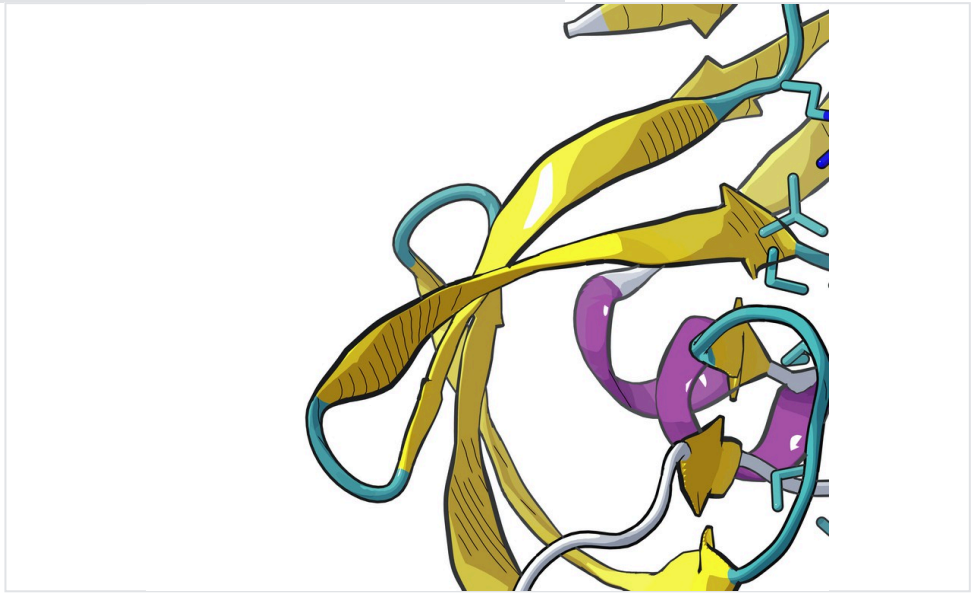
Shading in colour, not in black.

Light, mid, shadow: black ink over the colour (stage 4) against the shade colour (stage 5)





Stage 4: black ink at an opacity. The yellow strands turn olive-khaki in their shadow.
1HSG, crop · stage 4 final



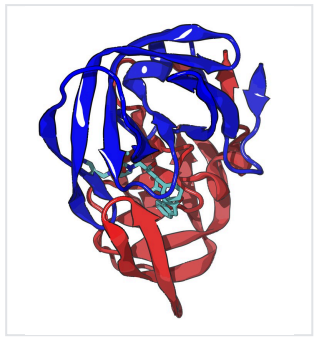
Stage 5: each band a deeper, warmer yellow; the teal coils go deep teal, the white coils cool slate.
1HSG, crop · stage 5 default

Every known defect, before and after.

FIXED · STAGE 2



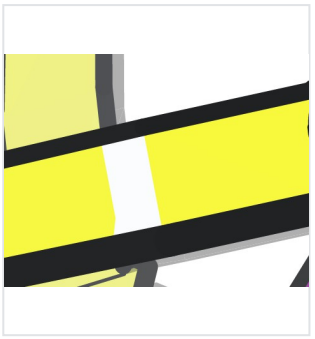
before



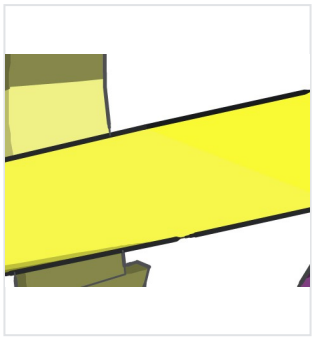
after

Lines too heavy

FIXED · STAGE 2



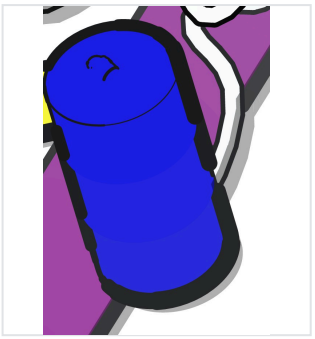
before



after

Seam: white notch in an arrow

FIXED · STAGE 2



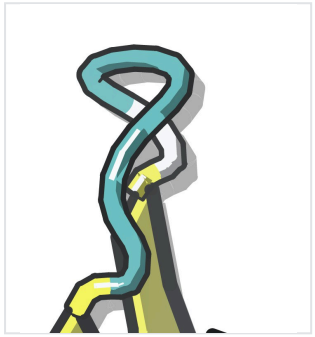
before



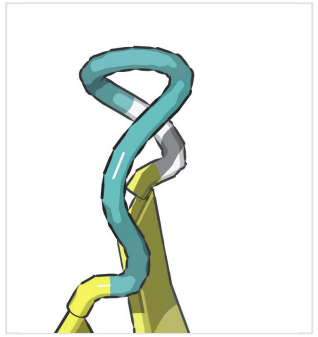
after

Banding across cylinders

FIXED · STAGE 2



before



after

Sticker drop shadow

FIXED · STAGE 2



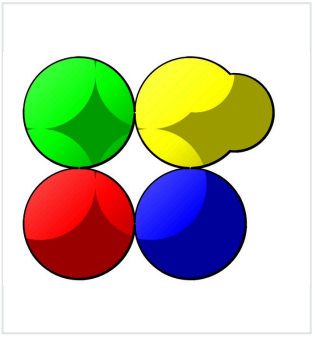
before



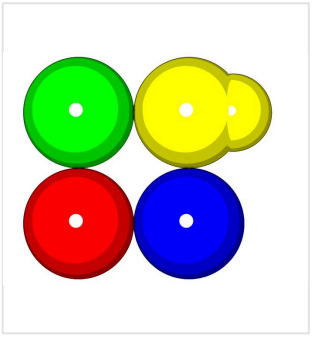
after

Stair-stepped bands, polygon glints

FIXED · STAGE 2



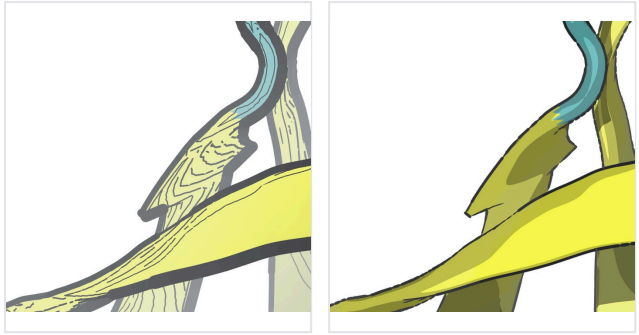
before



after

Stars on touching spheres

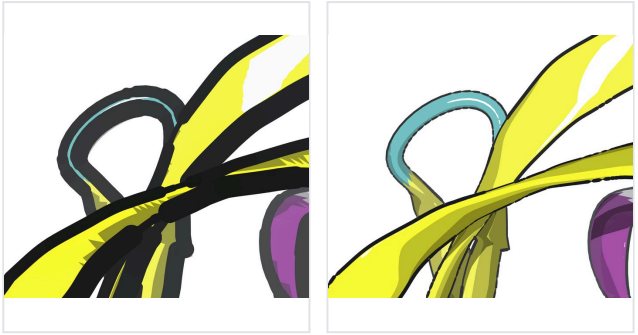
FIXED · STAGE 2



before after

Wood-grain hatching

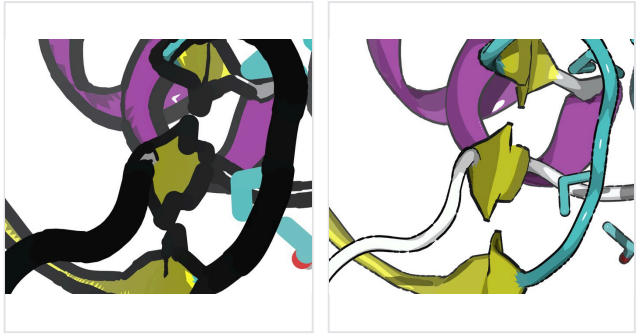
FIXED · STAGE 2



before after

Serrated band edges

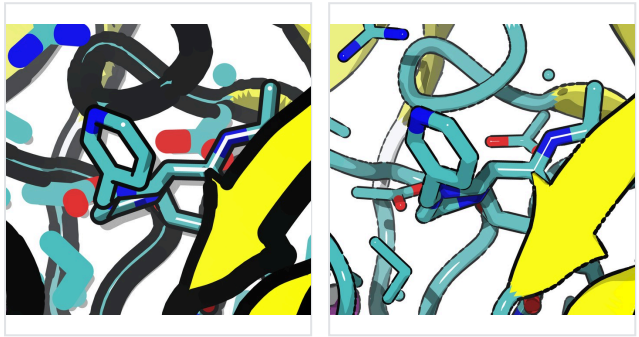
FIXED · STAGE 2



before after

Ticks and stray line pieces

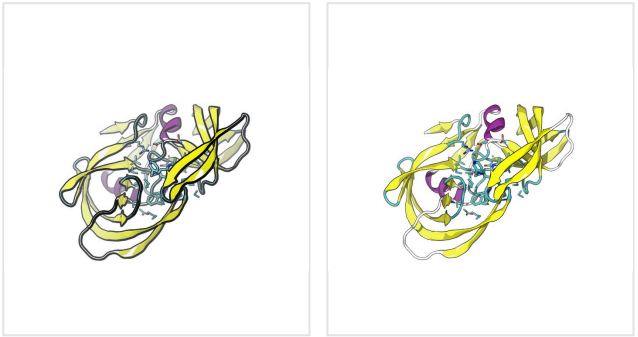
FIXED · STAGE 2



before after

Flat sticks

FIXED · STAGE 2



before after

Far shapes wash out (latest recipe)

OPEN · STAGE 4

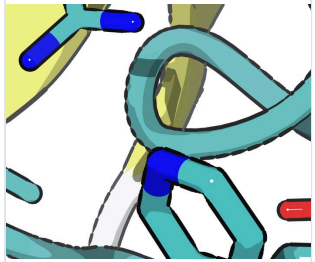


before after

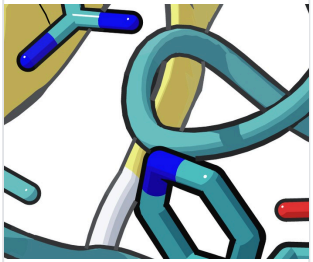
Transparent surface barely visible

What stage 3 fixed.

FIXED · STAGE 3



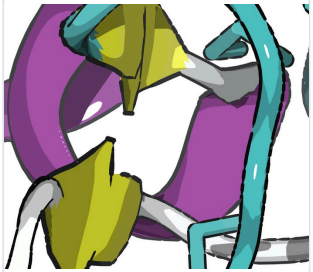
before



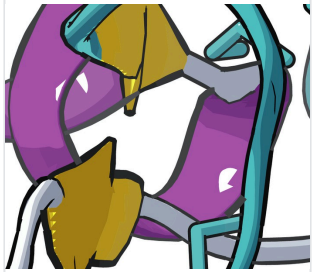
after

Dashed outlines. Stage 2 cut each outline into many short pieces (decided by eye rays, stepped widths, dropped ticks, tapered ends). Stage 3 chains them into one smoothly varying line per edge.

FIXED · STAGE 3



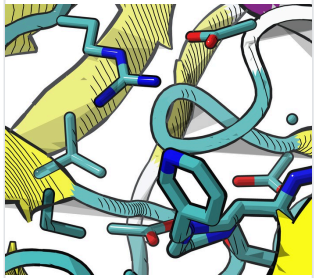
before



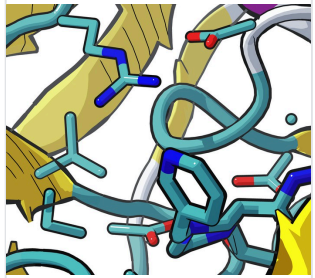
after

Stripe-like bands on loops. Bands from VMD's near-frontal light, smoothed on a pixel grid, broke into stripes on coils. Now from the upper-left key light, computed on the surface with consistent normals.

FIXED · STAGE 3



before

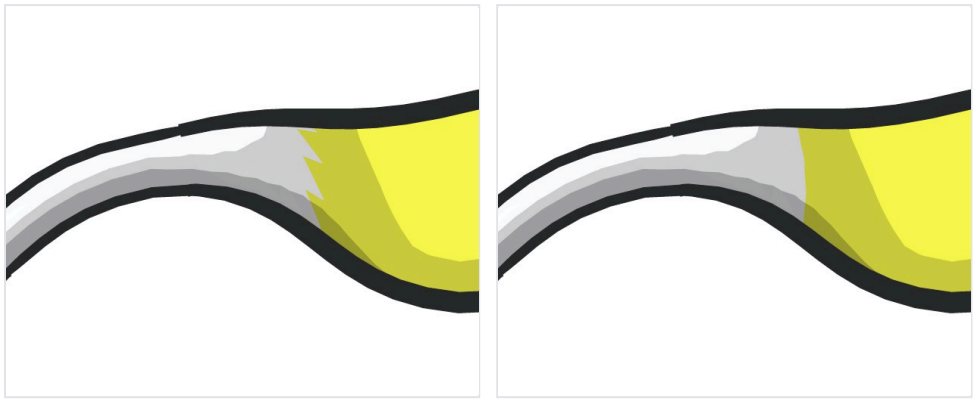


after

Wedges across the picture. Found during stage 3: merging tone pieces closed broken loops with a straight step, drawing grey wedges over the background. Such loops are now dropped and the pieces are built so they close.

What stage 4 fixed.

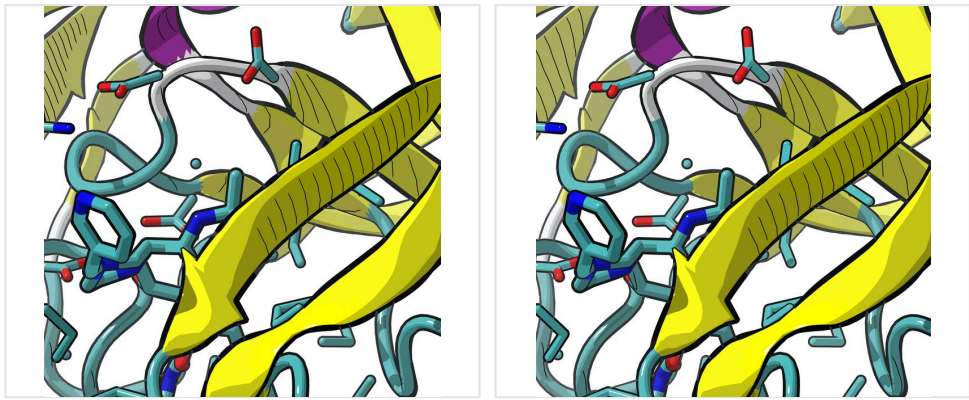
FIXED · STAGE 4



before | after

Torn colour boundaries. VMD colours each corner of the strip; stage 3 gave each triangle its corners' majority colour, a saw-tooth. Now each triangle is split where the colours meet, half-way between VMD's rings of points: one clean cut across.

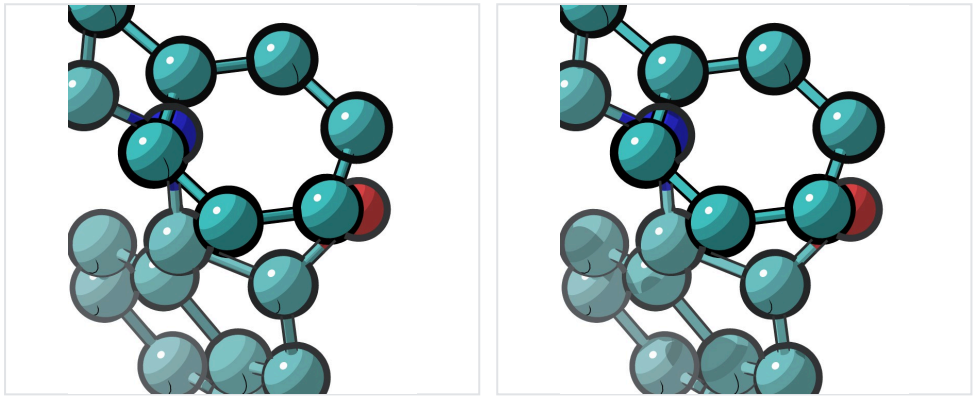
FIXED · STAGE 4



before | after

Busy hatching. No strokes where a mesh is seen nearly edge-on (they bunched into scraps on the helix's inner faces); small and far shapes get every other stroke, the tiniest none.

FIXED · STAGE 4



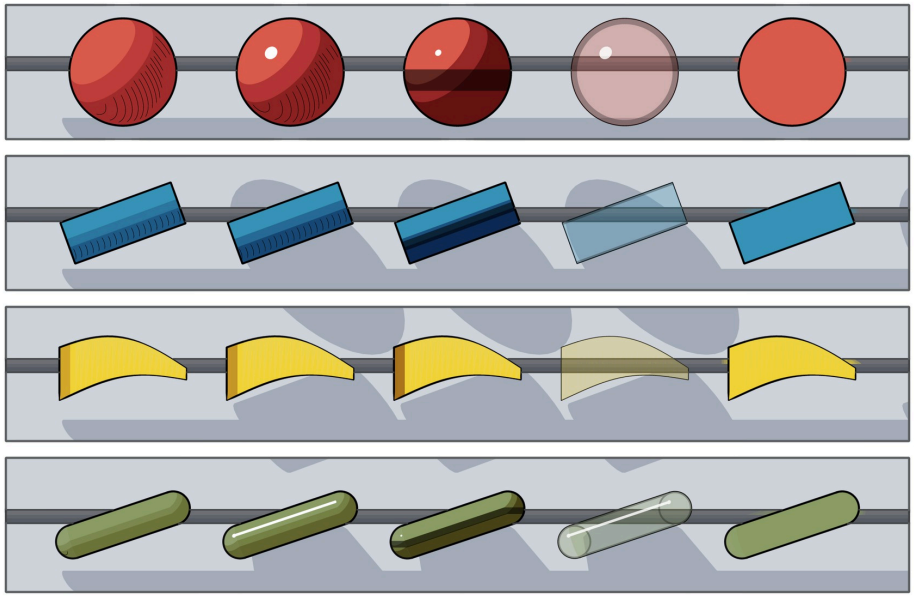
before | after

Shadows on balls. Stage 3 left out the shadows on balls whose outline the region engine could not close. Now built directly: the occluder's outline as the light sees it, cut to the ball's lit cap, lifted onto the ball, cut to its front half.

Defects that are not a crop.

DEFECT	EVIDENCE	STATUS
Materials and lights are ignored	Stage 4: five materials read from VMD's (or set per group), five lighting presets, VMD's first light with light=scene. Still ignored: AO, depth-cue variants, VMD's outline materials (Goodsell, Edgy)	MOSTLY FIXED stage 4
Black shading muddies colours	stage 4: tone bands were black ink at an opacity, so yellow went olive and the picture greyed. Stage 5: each band is a solid, deeper, hue-shifted colour made from the fill (page 7)	FIXED stage 5
Fill lost where a shape folds over itself	found in stage 5 on the ring model, then on a ribbon's curl in a test frame, where it had hidden since stage 2 (grey bands over the empty hole looked like a white ribbon; coloured bands gave it away). Merging one colour's pieces into one outline dropped loops it could not close, and with them a stretch of fill. Now such a colour keeps its pieces unmerged (that test frame 1.28 → 1.42 MB). Also: triangles of one colour lying wholly in front of each other now hide each other	FIXED stage 5
Fog steps show on large flat faces	the fog changes in steps per shape; on a model's large flat faces a step is a visible seam. Models get no fog by default (`fog=` turns it on)	WORKED AROUND stage 5
Large scenes are too big	stage 2: 1.4–40 MB for 400×304 crowds, 30.6 MB for the 292 000-atom membrane: every short line piece its own path. Stage 3 joins the pieces into one line per edge; stage 4 caps hatching and tones on small and far shapes (156 MB for the 11 large scenes, from 192). Stage 5's shadows-on default adds cast shadows: the 96-protein crowd 19 → 25 MB (`shadows=off` gives 19 MB; tiny shapes take no shadows, at most 48 per ball). The 292 000-atom membrane: 186 → 133 s alone (target 60 s)	PARTLY OPEN
Refusal without a picture	perspective with the camera inside an atom: an error, no fallback	OPEN stage 2–4
Groups not named by layer	the file's groups are now fills (one per scene group), tones, shadows, hatching, highlights, lines	FIXED stage 2
The shadow switch is ignored	stage 4: shadows followed VMD's switch. Stage 5 (owner): Comic Ink draws cast shadows by default; `shadows=scene` follows VMD's switch, `shadows=off` turns them off	FIXED stage 5
Cast shadows on tubes partial	balls (stage 4) and triangles are exact; on tubes the region engine still cannot close some loops, which are left out	OPEN after 1.0
Contact accents removed	stage 4: rebuilt from the geometry on meshes (ribbons, surfaces); not yet between two sticks or balls	FIXED stage 4 (meshes)
Mesh self-shadowing	tried in stage 4: +2.5 % on a test frame but jagged shards on the helix's inner faces; left out (owner, stage 5). Visible on the knot model: it casts no shadow on itself	LEFT OUT
`render SVG` ignored styles	VMD's command always gave the old VMD-matching output	FIXED stage 1

Five materials, drawn in ink.



The same ball, tube, ribbon and stick in each material (left to right: matte, plastic, metal, glass, emissive), with a dark bar and a pale strip behind each row to show what glass lets through and where each casts its shadow.

`stage 4 chart · made-up geometry, scene written by the stage 4 chart script · --style mat=Emissive:emissive`

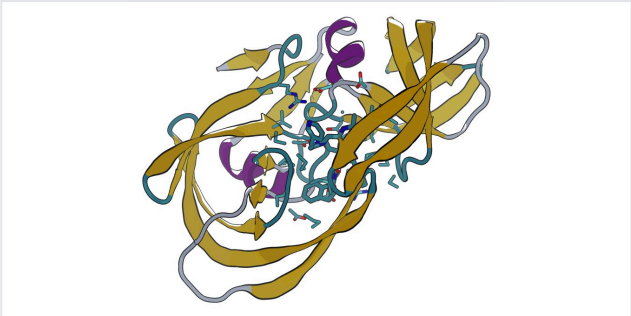
MATERIAL	READ FROM VMD WHEN	HOW IT IS DRAWN
matte	VMD Diffuse, AOChalky (no highlight)	no highlight; bands 0.7 × as dark
plastic	VMD Opaque, Glossy, HardPlastic (the rest)	the default look: two bands, a round highlight, hatching
metal	a mirror (RTChrome), or diffuse ≤ 0.4 under a highlight ≥ 0.3 (BrushedMetal, MetallicPastel, Steel)	bands 1.7 × as dark; a dark band where N·up is −0.35…0.05 (the horizon it mirrors); a small sharp highlight; no hatching
glass	opacity below 0.99 (Transparent, Glass, Ghost)	a tint (never lighter than luminance 0.75, so a white surface still shows) with what lies behind readable; on balls and tubes a darker rim where N·eye < 0.45 and a highlight; no shadows taken
emissive	the recipe only (VMD has no glowing material; its unlit Points and Lines stay plastic)	no bands, no hatching; three rings of its own colour around it, 5 px apart, each 22 % (vector strokes, no filter)

Per group in the recipe: `mat=metal` (all) or `mat=Rep[1]:glass;Licorice:metal` (the first part of a group name that matches wins).

One scene, five lights.



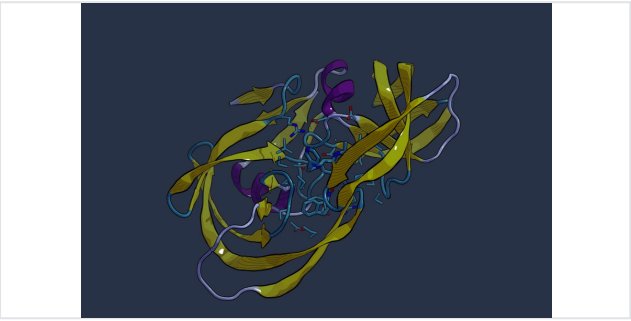
Studio (default)
`default · 1HSG`



Rim
`--preset rim · 1HSG`



Low
`--preset low · 1HSG`



Night
`--preset night · 1HSG`



Scene (VMD's light)
`--preset scene · 1HSG`

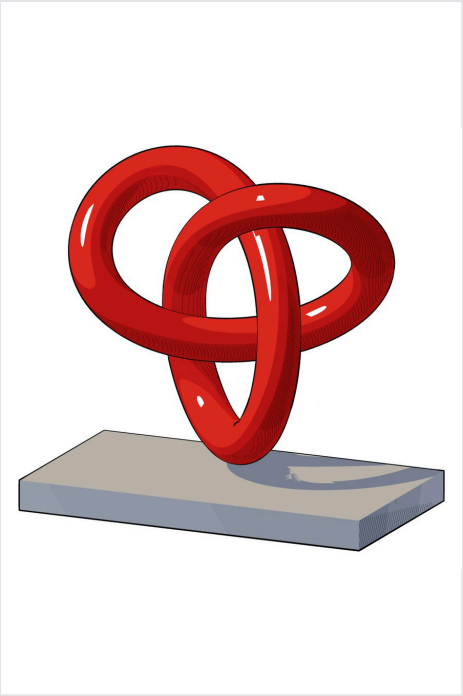


Studio, following VMD's shadow switch (off in this scene)
`--style shadows=scene · 1HSG`

PRESET	LIGHT
studio	the default: key from the upper left, 45° up, slightly in front
rim	from behind and above: the forms dark, a white lit rim where they turn to the light; no hatching
low	from below at the front
night	a cool key from the upper right over a dark blue ground; colours × (0.60, 0.66, 0.85), far shapes fade into the ground
scene	VMD's first light: a directional one as it is; a point light as seen from the scene's middle (one direction for the whole picture)

Cast shadows are drawn by default (owner, stage 5); `shadows=scene` follows VMD's shadow switch (on when Tachyon's Shader_Mode is Full), `shadows=off` turns them off.

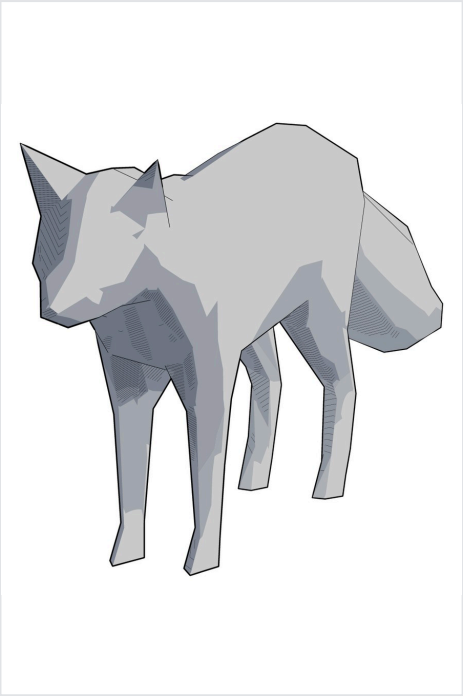
Any scene, the same ink.



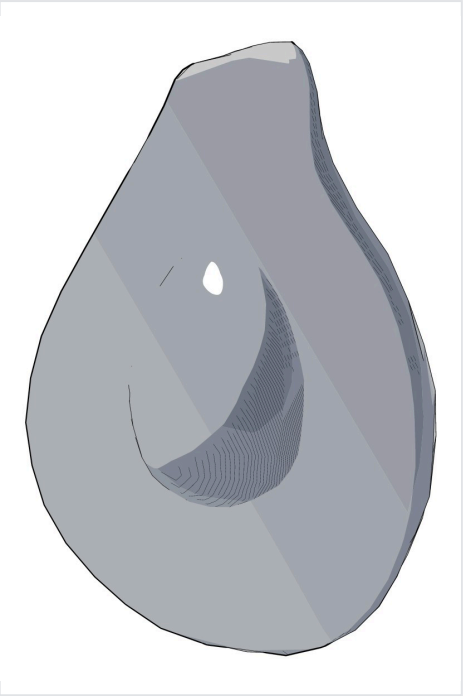
A trefoil knot on a plinth: an OBJ model, two materials from its .mtl. A mesh does not shadow itself (page 8).
`examples/meshes/knot.obj`



A glTF still life: glass, brass, plastic and wood, from glTF's base colour, metallic, roughness and alpha.
`examples/meshes/still_life.glb`



A third-party model: the Khronos Fox (CC0 / CC BY 4.0, PixelMannen, tomkranis, AsoboStudio, scurest), its rest pose. Its colour is only in a texture: drawn light grey.
`examples/meshes/fox.glb`



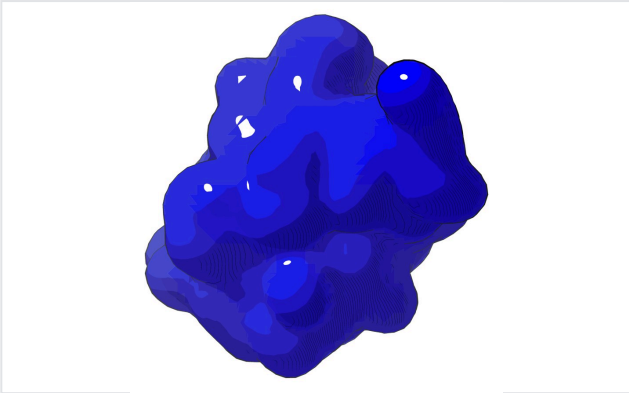
The Khronos Avocado (CC0): half an avocado, 6 cm across in metres, the stone raised on the cut face. Drawn the same at 1000 times its size.
`Khronos Avocado.glb`

OBJ (with .mtl) and glTF 2.0 (.gltf, .glb) are read into the same internal scene as VMD's files; nothing in the drawing knows what a molecule is. Smooth triangles where the file has normals, flat ones where it has none; each object becomes a group the recipe's `mat=` can pick. The model is turned into Tachyon's left-handed space (z mirrored, as VMD does: drawn the right way round, never as its mirror image) and scaled to VMD's size, so a model of any size draws the same. The file's camera is used if it has one, otherwise an orthographic camera frames the model from the front, a little from the right and above, and Comic Ink's key light goes to the side the seen faces lean towards, so the side you see is lit. Textures, skins, animation, points and lines are reported once and left out; a colour only in a texture becomes light grey. Models get no depth fog by default. The knot and the still life were made for this project by `scripts/make_meshes.py` (CC0); the fox and the avocado are Khronos glTF samples (THIRD-PARTY-NOTICES.md).

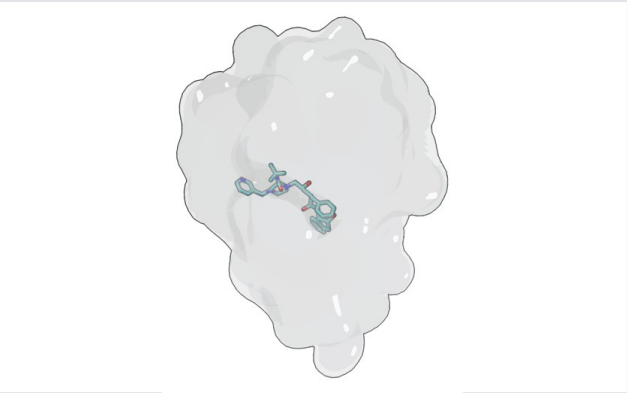
Version 1.0, as it ships.



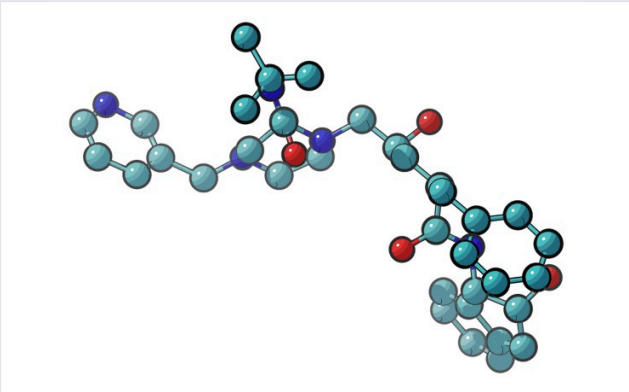
Whole-protein cartoon (1HSG)
1.0.0-rc1



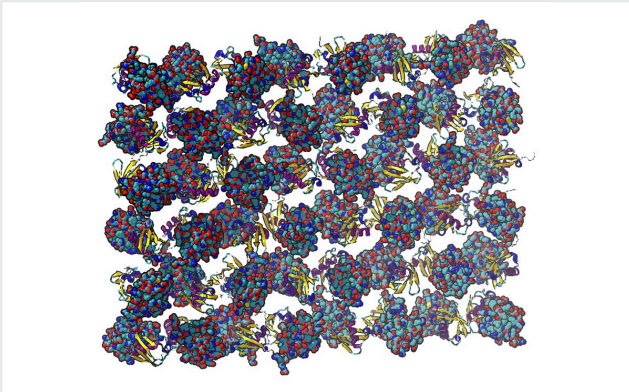
Surface (QuickSurf, 1UBQ)
1.0.0-rc1



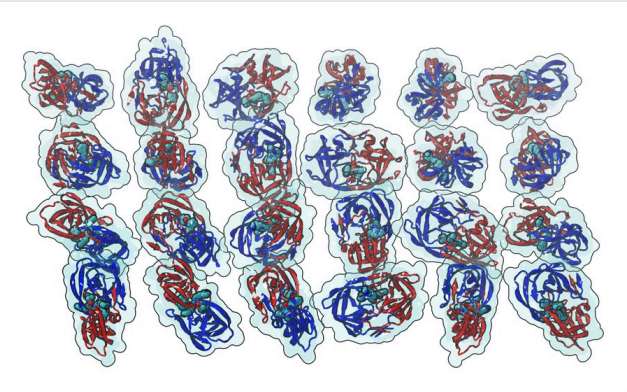
Transparent surface over a ligand (1HSG)
1.0.0-rc1



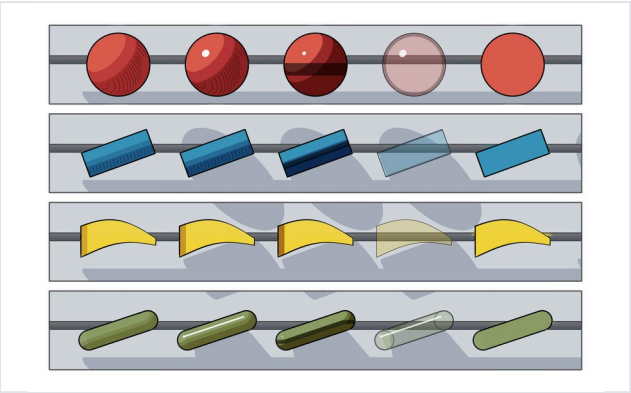
Ball-and-stick (indinavir)
1.0.0-rc1



Large scene: 96 ubiquitins
1.0.0-rc1



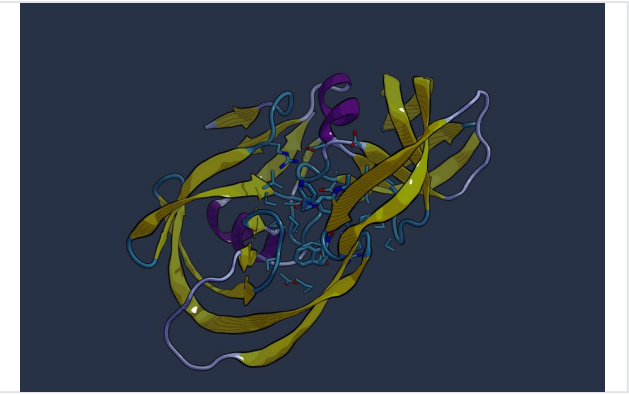
Large scene: a slab of 24 proteases under glass
1.0.0-rc1



Material chart
1.0.0-rc1



--preset noir
1.0.0-rc1



--preset night
1.0.0-rc1